

INPUT

M METAMODEL

DCL

CSL

FSL

Predicates C1–C9

Traces + format[1]

Cross-layer graph predicates

S SCENARIO

Housing subsidy (CSOK)

3 credentials

3 issuers

Cross-credential deps.

Value constraints

G GOVERNANCE CONSTRAINTS

eIDAS ARF

GDPR 5(1)(c)

W3C VCDM 2.0

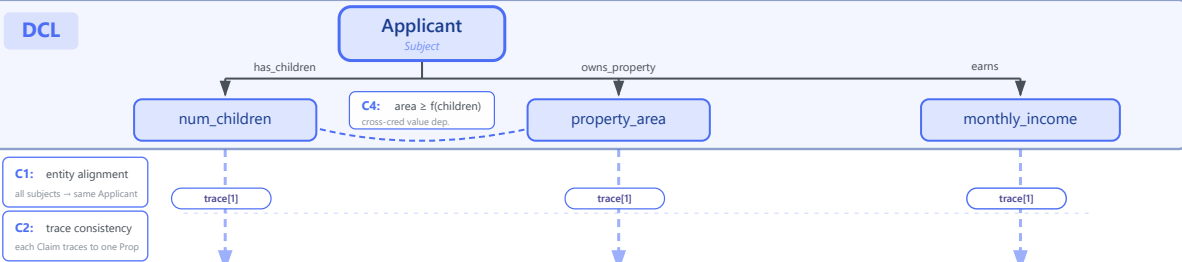
Format mandates

Privacy reqs.

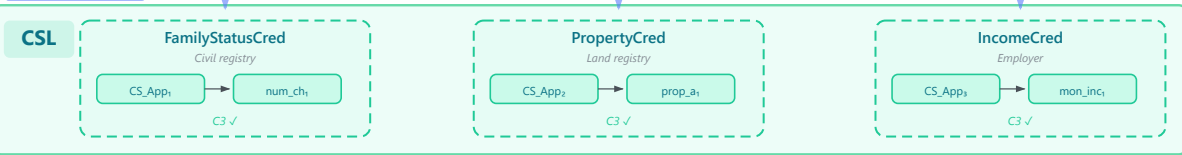
Data model conformance

MODEL

DCL



CSL



C5:

eIDAS mandate

EU wallets → SD-JWT-VC

format[1]

C6:

pred proof req.

GDPR → income needs ZKP

format[1]

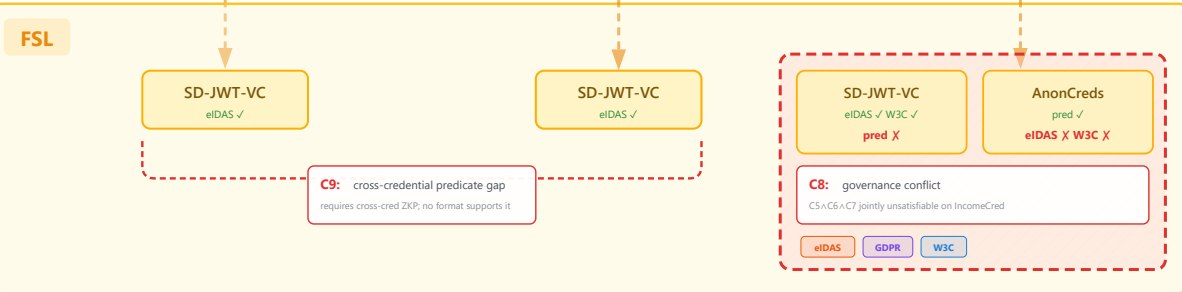
C7:

VCDM conform.

W3C → AnonCreds fails

format[1]

FSL



RESULT

1 VERIFY

SAT

design is consistent

UNSAT

constraints conflict

2 IDENTIFY ERRORS

► **governance_conflict**

(IncomeCred)

► **cross_cred_pred_gap**

(area, children)

3 GENERATE

Relax C6 (GDPR) → 2 configurations

